Symmetry-Safe Adaptive Noise Scheduling for E(3)-Equivariant Diffusion Training

EECE 571F Project Presentation

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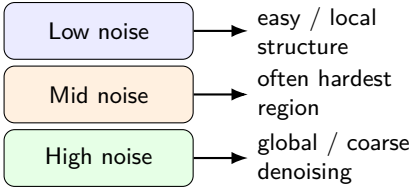
1. Motivation and Problem

- Diffusion models train over many noise levels, but most implementations use a fixed timestep schedule.
- Different timestep regions can have very different difficulty, variance, and optimization value.
- In 3D molecular diffusion, this matters even more because the model must denoise while preserving **E(3) symmetry**.



Core question

Can we spend more training effort on hard timestep regions *without* changing the baseline objective?

Why this matters

If successful, we learn whether training compute can be allocated more efficiently in equivariant molecular diffusion.

2. Related Work and Project Gap

Area	Takeaway
Equivariant molecular diffusion	Strong baselines already exist for 3D molecule generation; we build on them instead of inventing a new backbone.
Diffusion scheduling / weighting	Prior work suggests timestep choice matters for optimization efficiency and variance.
Our project gap	Very little work studies **symmetry-safe adaptive timestep allocation** for **E(3)-equivariant** molecular diffusion.

Our positioning

This is a **training-time optimization project**, not a new molecule generator architecture.

- Baseline: public EDM-style equivariant molecular diffusion codebase.
- Goal: improve how timesteps are sampled during training.
- Constraint: preserve the original objective in expectation and use symmetry-safe hardness signals.

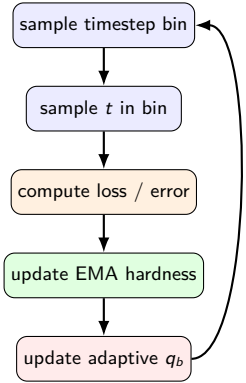
3. Main Idea: SAANS

- Partition timesteps into bins $I_1, \ldots, I_B$.
- Track a detached hardness estimate S_b for each bin.
- Define an adaptive distribution:

$$q_b \propto p_{0,b}(\varepsilon + S_b)^{\alpha}$$

- Mix with the baseline for stability and full support.
- Importance-weight each sampled bin by

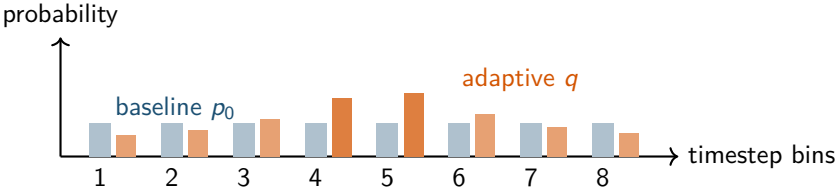
$$w_b = \frac{p_{0,b}}{q_b}$$



Key design constraint

Hardness estimates should be **symmetry-safe**: they should not change under arbitrary rotations or translations.

4. Draft Main Figure: Reallocating Timestep Mass



Interpretation

Illustrative draft figure: the baseline schedule is fixed, while SAANS shifts probability mass toward bins that appear harder under training diagnostics.

- Final paper figure should show how q_b evolves over training.
- We also want matching plots for hardness EMA, occupancy, and per-bin loss variance.

5. Planned Experiments and Current Status

Planned QM9 study

- Baseline EDM scheduler
- Full SAANS
- Adaptive without importance weighting
- Hardness variants and scheduler sweeps

What we want to learn

- Which timestep regions are hardest?
- Does adaptive allocation improve optimization efficiency?
- Is importance weighting necessary?
- Does symmetry-safe hardness matter?

Current project status

- Public EDM baseline integrated
- QM9 pipeline working locally
- Smoke tests for baseline and SAANS completed
- Short-run comparison pipeline built

Important caveat

Current numbers are only **smoke-test engineering checks**, not publication-grade results.

6. Takeaway

- We study **adaptive timestep allocation** for **E(3)-equivariant molecular diffusion**.
- The project contributes a **training-time scheduler** rather than a new backbone.
- The key idea is to reallocate probability mass toward hard timestep regions while preserving the baseline objective through importance weighting.
- The implementation scaffold, baseline integration, QM9 pipeline, diagnostics, and short-run SAANS tests are already in place.
- The next step is to run longer GPU experiments and evaluate whether SAANS improves efficiency and generation quality.

Questions?